

Experiment design and literature review

Sofia Razou

July 2026

1 Literature insights

Design, Control, and Evaluation of a Family of Kinesthetic Haptic Interfaces

This thesis examines the aspects of performance for haptic interfaces with force feedback (kinesthetic, rather than tactile), especially exhibiting interesting haptic definitions, control algorithms, the issues the parameters the haptic device pose for the transparency and stability of the system, alongside measures to alleviate the aforementioned issues.

Pivotal Definitions

1. Backdrivability: In unconstrained motion, no force generated by the natural device dynamics should be evident to the operator.
2. (apparent) Stiffness: Measure of virtual elasticity provided by the device's interface for rendering impedances in virtual environments, at the human interface .

The control stiffness can be enhanced with increasing the physical damping or sampling rate.

Coulomb friction assists in higher stiffness and stability, however from a point onwards presents the operator with input-state stability bounds.

Issues to be tackled in the framework of the CAPT Motor as a haptic force-feedback device

1. The application of active feedback control makes the a priori estimation of achievable performance improvement a real challenge.
2. Need for crisp haptic fidelity and transparency at the render of hard stops.

Performance measures

1. Mechanical Hardware features:

- (a) Performance criterion: provided by the device's hardware specifications
- (b) Dexterous workspace, free of singularities, DOF number, range of Eulerian angles etc.
- (c) Output capability: max current, torque output and angle range, max payload
- (d) Sensorial Capability: accuracy of sensors (here optical encoder, maybe accelerometers and insert latency for acceleration tracking, probabilistic model)

2. Closed-loop performances

- (a) Force precision: zero position, with no commanded output (control goal).

It is highly dependent on the capability of the control scheme to attenuate stiction effects. A good specification for idle force-precision is the consideration of the worst case figure in the entire workspace.

One must also take into account the smallest force error humans are able to perceive while interacting with such devices, which is estimated by subjective measures (like JND), being almost linear to the level of the commanded force.

- (b) Dynamic force precision: zero (near-zero) offset for angle commands of various acceptable ranges for the device's capabilities (clamped end-effector mode).

Dependent on the resonant frequencies of the device, however a good measure is the measurement of the device's response to sinusoidal force commands (really positive news for the CAPT Motor, since it functions by design with sinusoidal torque commands).

Usage of frequency response experiments for the quantification of distortion of the rendered force signal, due to the sensors' noise. (see stochastic processes that could be suitable).

- (c) Backdrivability: the ability of the haptic device to trace back the commanded path after it being executed.

This KPI is of grave importance for the emulation of impedances falling below the natural (Free-space rendering) impedance of the haptic interface (e.g. free space simulations).

Divide and conquer method for its evaluation: subdivide it with respect to the originating physical effects and then to define a meaningful measure for each contribution i.e. gyroscopic effects, frictions of all significant present kinds, maximum inertia. (like with the usage of hamiltonian decompositions of the system's equivalent graphion)

Take into consideration of the direction of the impedances, in terms of isotropy and cross couplings to take into account the several disturbance forces, to ensure naturality at the human interface.

Also, usage of frequency response to sinusoidal input.

- (d) Stiffness: the device's capability for the realistic render of rigid objects in addition to the force bandwidth determined by the maximum closed-loop stiffness. Usually it is derived with equation resolution with encoder-measurements.

Determination of actual Performance Indices

The following parameters for experimental design and evaluation are subject to either direct measurement or derived by analysis of an appropriately characterized device software model.

Two approach perspectives: one is measurement-based and the other is based on model formulation and prediction.

Pros of measurements over model-based evaluation: Direct measurement tends to be more reliable when the model presents many inaccuracies (the CAPT Motor has a really clear physical model).

Cons of measurements over model-based evaluation: The identification of worst case figures is hard to obtain with direct measurement. Additionally, overall measurements are gain-dependent so generalization of their conclusions might be off from the general reality.

Overall, it is reasonable to perform hardware experiments whenever necessary for data-acquisition but to make use to the fullest extent of the model-based performance estimation.

e.g. realistic gain estimation: by hardware experiments, coupled with a human in the loop (HIL).

Experiment design ideas

1. Benchmarking of the CAPT Motor against a conventional DC steering wheel motor (mount to it the same bearing as in the CAPT Motor, so as to retain as many of the non-significant parameters the same across the two platforms).
2. Assess quantively the transparency of the Motor's with accelerometer attachments to the operator's main force feedback-receiving end joints and in the Motor. This will allow potentially for a neural network based controller with inference to approximate the rendered impedance.
3. Human-in-the-loop with user evaluation studies and cross-reference them with those drawn from the inferred model.
4. Develop three main control schemes and use impedance and port-hamiltonian systems to evaluate the performance while maintaining stability and high-rendering ability: P, PD, Inertia -impedance based, Energy-based (less

conservative than the prior degenerated passivity controllers), port-hamiltonian framework with hamiltonian decomposition of the graph model. (add observers when needed).

5. Different virtual environments: virtual springs, inertia, damping, acceleration, deceleration, virtual lane assist.